Coverage-Qualified Evaluation-Contract Auditing for Aerial Vehicle Detection

Xinling Wen, Jiahao Zhang, Wenqi Liu, Yangxu Ren, and Yu Chen

***Abstract*—Aerial-detection scores change when a support rule alters the scored target and the treatment of detections near excluded objects. We introduce a coverage-qualified evaluation-contract audit comprising a coverage gate, support measurement, an ordered contract path, and common-coverage rank inference. The path $A \rightarrow F \rightarrow R \rightarrow S$ changes one evaluator factor at a time and attaches an exact TP/FP/FN ledger to target filtering, removed-object neutralization, and source-ignore handling. At 24 pixels, VisDrone, UAVDT, and AI-TOD retain 73.3%, 95.1%, and 9.7% of valid vehicle boxes; AI-TOD prediction coverage contains 66.7% of images and 28.2% of boxes. Across five VisDrone and four UAVDT full-coverage configurations, removed-object neutralization raises F_1 in all 18 support-rule replays. On 548 common VisDrone images, simultaneous max-t cluster intervals resolve three of four max-F_1 cells for Y11m, absolute AP25 for Y11m, and normalized AP50 for ASD; three cells remain unresolved. Retention-matched minimum-side and geometric-mean support preserve every directional point comparison.**

***Index Terms*—Aerial detection, annotation support, evaluation contract, coverage qualification, cluster bootstrap.**

I. Introduction

A detection score is interpretable only after its evaluation target and matching contract are fixed. This requirement is acute in aerial imagery: a common pixel cutoff can retain almost every vehicle in one source and remove most vehicles in another. Predictions near removed objects may then be counted as false positives or treated as neutral, while source-provided ignore regions add a separate decision. These choices enter the metric before detector outputs are compared.

Cross-dataset bias, heterogeneous label spaces, annotation reliability, and detector error taxonomies are well studied [1], [2], [3], [4]. Tiny-object procedures can also alter the prediction process itself [5]. This study instead measures evaluator effects after annotations and prediction caches are fixed, attributes those effects to explicit contract steps, and separates them from sampling uncertainty. Recent GRSL work likewise shows that validation conditions can materially affect remote-sensing conclusions [6], [7], but aerial-detection reports rarely connect coverage, support, contract semantics, and rank inference in one record.

Three weaknesses make an unstructured sensitivity sweep hard to defend. First, incomplete prediction coverage can silently change the evaluated population. Second, endpoint ranges do not identify whether movement follows target filtering, the treatment of removed objects, or native ignore regions. Third, a fixed-confidence F_1 comparison can confound detector ranking with score calibration, while image-wise resampling can ignore sequence dependence. We address these weaknesses without training another detector.

Our contributions are fourfold. 1) We define a coverage-qualification gate that separates full-dataset annotation support from prediction-covered metric inference. 2) We introduce an ordered evaluation-contract path with an exact count-transition ledger. 3) We replay the complete path on nine full-coverage configurations from VisDrone and UAVDT, with three coverage-conditioned AI-TOD resolution controls [8], [9], [10]. 4) We qualify the named VisDrone pair with simultaneous sequence-cluster intervals, a localization-by-metric control, and two retention-matched support definitions.

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II. Coverage-Qualified Contract Audit

A. Gate 0 and support coordinates

We use VisDrone2019-DET validation, the UAVDT test split represented in COCO format for a common evaluator, and AI-TOD validation. The available AI-TOD reconstruction includes xView-derived imagery [11]. Vehicle mappings are VisDrone {car, van, truck, bus}, UAVDT {car, truck, bus}, and AI-TOD {vehicle}. Raw coverage is established from prediction image names *before* class filtering; a covered image with no retained vehicle prediction is an empty prediction, not a missing observation.

Gate 0 records the annotation denominator, raw coverage, and covered–noncovered differences before any metric is interpreted. Complete artifacts can enter all later layers. An incomplete artifact may enter support diagnosis and a clearly conditioned metric replay, but not a full-benchmark rank claim. We compare AI-TOD coverage groups by boxes per image, support, aspect ratio, and normalized nearest-neighbor spacing; standardized mean differences and Wasserstein distances are released.

Formally, let $\mathcal{I}_s$ be the declared annotation-image set for source s and $\mathcal{P}_k$ the image identifiers present in candidate artifact k. Its raw coverage is

$$q_{sk} = \frac{|\mathcal{I}_s \cap \mathcal{P}_k|}{|\mathcal{I}_s|}. \tag{1}$$

Annotation support is estimated on all of $\mathcal{I}_s$ and is independent of predictions. A metric endpoint for k is a full-source estimand only when $q_{sk} = 1$; otherwise it is written conditionally on $\mathcal{J}_{sk} = \mathcal{I}_s \cap \mathcal{P}_k$. A pairwise rank estimand uses the explicit common set $\mathcal{J}_{sk\ell} = \mathcal{I}_s \cap \mathcal{P}_k \cap \mathcal{P}_\ell$. This separation prevents a high-coverage artifact from silently redefining the benchmark denominator and prevents unmatched image sets from entering a detector comparison.

For a box (w, h) in image (W, H), absolute and normalized max-side support are

$$s_a = \max(w, h), \qquad s_n = \max\left(\frac{w}{W}, \frac{h}{H}\right). \tag{2}$$

We use the prespecified support grids $s_a \in \{16, 20, 24, 28, 32, 40, 48, 64\}$ pixels and $s_n \in \{.005, .0075, .010, .015, .020, .030\}$; 24 pixels and .015 are the headline interior values. Under uniform resizing by $\alpha > 0$, $s'_a = \alpha s_a$ whereas $s'_n = s_n$. Thus normalized support is resize invariant and absolute support is scale covariant. This property does not imply semantic equivalence across sources; ontology, acquisition geometry, cropping, and density can still differ.

Proposition 1 (scale behavior). Under the uniform transformation $(w, h, W, H) \mapsto (\alpha w, \alpha h, \alpha W, \alpha H)$ with $\alpha > 0$, normalized max-side support is invariant and absolute max-side support is covariant. This follows directly because $\max(\alpha w, \alpha h) = \alpha s_a$, whereas both factors of α cancel in s_n. The proposition motivates reporting both coordinates; it does not make a shared normalized threshold a claim of semantic equivalence.

B. Ordered contract-path decomposition

Let G be mapped valid boxes, $G_t \subseteq G$ the boxes retained by support rule t, $D_t = G \setminus G_t$ the removed boxes, and H the source-provided ignore regions. We declare four contracts as ordered pairs of scored valid and neutral regions:

$$\begin{aligned} \mathcal{C}_A &= (G, \varnothing), & \mathcal{C}_F &= (G_t, \varnothing), \\ \mathcal{C}_R &= (G_t, D_t), & \mathcal{C}_S &= (G_t, D_t \cup H). \end{aligned} \tag{3}$$

A scores all valid boxes with ignore protection off. F filters the target and leaves detections on removed objects eligible as false positives. R keeps the same filtered target but makes removed objects neutral. S then adds source-provided ignores. In every contract, detections are processed by descending score, retained valid GT is matched first, and only an unmatched detection can be neutralized. Ignore overlap follows the COCO crowd denominator, intersection over prediction area, at .5 [12].

For metric m at rule t, define adjacent path changes

$$\begin{aligned} \delta_{\text{target}} &= m_F - m_A, \qquad \delta_{\text{removed}} = m_R - m_F, \\ \delta_{\text{source}} &= m_S - m_R. \end{aligned} \tag{4}$$

They satisfy the exact path identity

$$m_S - m_A = \delta_{\text{target}} + \delta_{\text{removed}} + \delta_{\text{source}}. \tag{5}$$

The associated ledger records the exact ΔTP, ΔFP, and ΔFN at each step. Because F_1 is nonlinear, these are path-specific evaluator transitions, not order-free causal effects. The deterministic contract range $C_m(t) = \max_{c \in \{A,F,R,S\}} m_c(t) - \min_c m_c(t)$ remains a compact summary, but it is not a confidence interval.

The path also imposes implementation invariants. Each detection can match at most one scored valid box; scored-valid matching precedes neutralization; and a detection neutralized by D_t or H is neither TP nor FP. For micro-$F_1 = 2\mathrm{TP}/(2\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$, the ledger is regenerated from image-level integer counts and must close both the count totals and the endpoint identity above. These checks distinguish a semantic contract change from accidental rematching or denominator drift.

C. Metric- and cluster-qualified rank protocol

The path replay uses five VisDrone configurations on 548/548 images and four UAVDT configurations on 305/305 images. The VisDrone pool contains B-640, B-1280, Y11m, ASD, and Y11n-P2; UAVDT contains baseline inference at 640, 960, and 1280 pixels plus 960-pixel tiling. Three AI-TOD resolution slices traverse the same path on their common 1,869-image covered subset and remain ancillary. Every configuration uses confidence and valid-match IoU .25. At terminal contract S, the VisDrone display reports micro-F_1 at confidence .25, max-F_1 over .10–.50, and AP50 with `maxDets`=2000. For Y11m–ASD, a 2×2 control crosses valid-GT IoU $\{.25, .50\}$ with max-F_1 and AP. Neutralization remains fixed at intersection-over-prediction .5.

VisDrone filenames identify 76 sequences. A paired cluster replicate samples 76 sequence identifiers with replacement, includes every image in each sampled sequence, applies identical multiplicities to both configurations, and recomputes pooled TP/FP/FN or pooled-detection AP. We use 10,000 common replicates. Single-step max-t intervals provide simultaneous 95% coverage for the eight max-F_1/AP comparisons formed by two supports, two valid-match IoUs, and two metrics. Image-paired results remain a supplementary control. An independent detection-record AP50 implementation is checked against COCOeval at the two candidates and two headline rules.

The ontology mappings, image denominators, support grid, contract order, candidate cohorts, named pair, AP cap, sequence definition, bootstrap seed, and replicate count are stored as machine-readable records. The VisDrone extension was frozen at public commit `11a7ca7` on 17 July 2026, 08:53:06 UTC. A second record fixed the cross-configuration cohort and input hashes at 09:34:40 UTC before replay. These are reproducibility freezes; the pair and headline rules originated in earlier audits. Minimum-side and geometric-mean-side thresholds use valid GT alone to match the max-side retained count.

III. Results

A. Coverage and support qualification

Figure 1 shows full coverage for VisDrone (548/548) and UAVDT (305/305), but 1,869/2,804 images for AI-TOD. At 24 pixels, full-source retained support is 73.3%, 95.1%, and 9.7%; at .015 it is 89.8%, 97.4%, and 81.0%. Normalization narrows the three-source headline spread but does not remove it.

AI-TOD's missingness is support selective. Its covered images are 66.7% of the validation images but contain

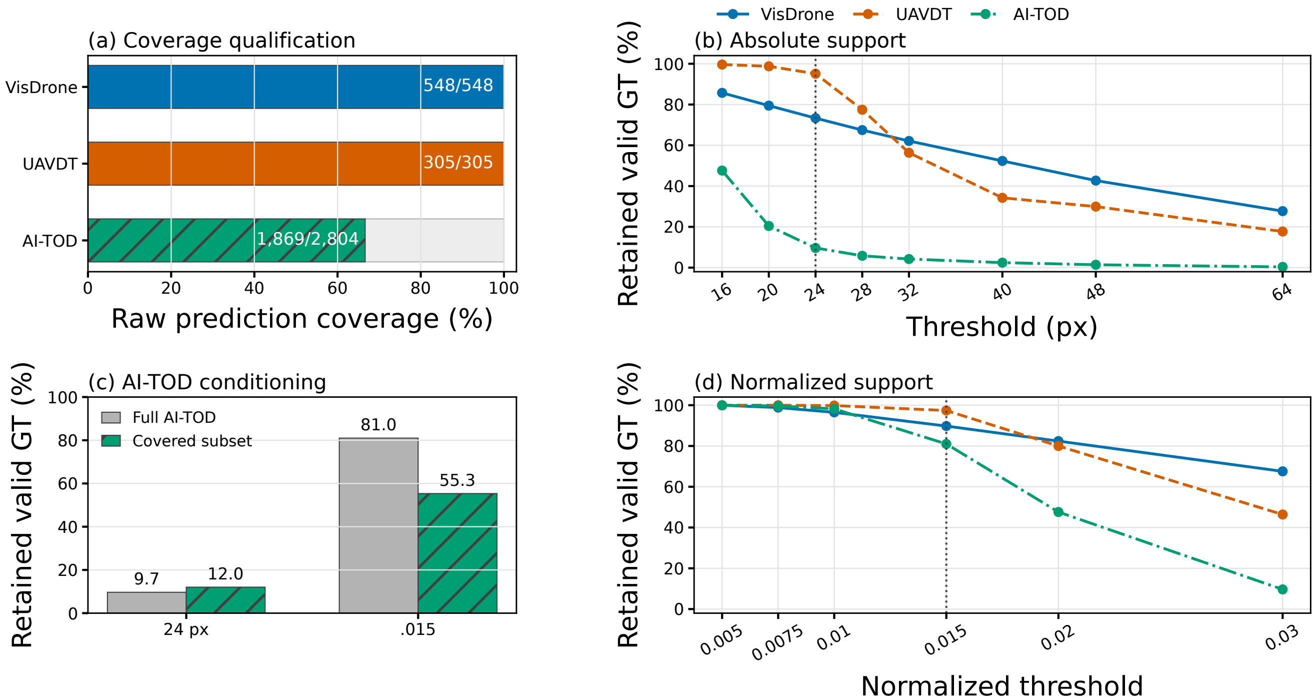

Fig. 1. Coverage qualification precedes metric interpretation. (a) Raw prediction coverage. (b) Full-source absolute retained-support curves. (c) AI-TOD's covered images contain 28.2% of its vehicle boxes; its full-source and covered-subset headline support therefore differ. (d) Full-source normalized retained-support curves. Percentages are pooled valid-GT fractions computed before prediction matching.

16,900/59,904=28.2% of vehicle boxes. Covered and noncovered images average 9.0 and 46.0 boxes; their medians are 0 and 21. The covered-subset retained fraction is 12.0% at 24 pixels and 55.3% at .015, versus 9.7% and 81.0% on the full annotations. The main analysis uses AI-TOD for coverage and full-annotation support qualification. Its covered-subset metric replay appears in the supplement.

The qualification is not triggered by coverage rate alone. Covered versus noncovered images have a boxes-per-image standardized mean difference of $-.673$; at box level, max-side Wasserstein distances are 2.96 pixels and .00370 in normalized units. Consequently, the uncovered part is neither treated as empty prediction nor repaired by weighting. The audit preserves two separate quantities: full-annotation support and covered-subset contract response.

B. Contract-path attribution

Figure 2 tests whether contract attribution survives a change of detector configuration. At 24 pixels, target filtering lowers F_1 for all five configurations by .031–.093. Removed-object neutralization then adds .067–.093, and source ignores add .009–.028. The resulting contract ranges span .076–.119. The exact ledgers close separately for every candidate.

At normalized .015, target filtering is configuration dependent: $A \rightarrow F$ ranges from $-.019$ to $+.025$. The next two steps remain positive for all five configurations, and the contract ranges narrow to .034–.049. The mechanism is stable across the pool even when the first transition changes sign.

Four full-coverage UAVDT configurations provide an independent source-level replay. Their contract ranges are .0030–.0147 at 24 pixels and .0018–.0041 at .015. Removed-object neutralization raises F_1 in every UAVDT configuration by .0029–.0067 and .0010–.0027, respectively. Across the nine full-coverage configurations, this transition is positive in all 18 rule replays. Source-ignore handling is active only in VisDrone, which separates a general removed-object effect from source-specific ignore metadata.

Retention-matched support controls preserve this structure. Minimum-side and geometric-mean-side rules yield absolute contract-range spans of .067–.115 and .073–.119, respectively; their normalized spans are .032–.047 and .034–.049. Across all 30 candidate–rule combinations, $F \rightarrow R$ and $R \rightarrow S$ are positive. The support functional changes the magnitude while preserving the two neutralization effects.

C. Metric- and sequence-qualified ranking

At the fixed confidence, pointwise sequence intervals favor Y11m in all four support–IoU cells. The simultaneous family treats max-F_1 and AP as the eight primary comparisons. Max-F_1 favors Y11m at absolute IoU .25 [.0097,.0230], absolute IoU .50 [.0076,.0207], and normalized IoU .25 [.0043,.0155]. Its normalized IoU .50 interval [$-.0010$,.0117] is unresolved.

Figure 3(b) separates score integration from localization under family-wise coverage. AP favors Y11m for absolute IoU .25 [.0033,.0129], remains unresolved for absolute IoU .50 [$-.0046$,.0072] and normalized IoU .25 [$-.0061$,.0051], and favors ASD for normalized IoU .50 [$-.0199$,$-.0022$]. Minimum-side and geometric-mean-side controls preserve the sign of every max-F_1 and AP point difference across the 2×2

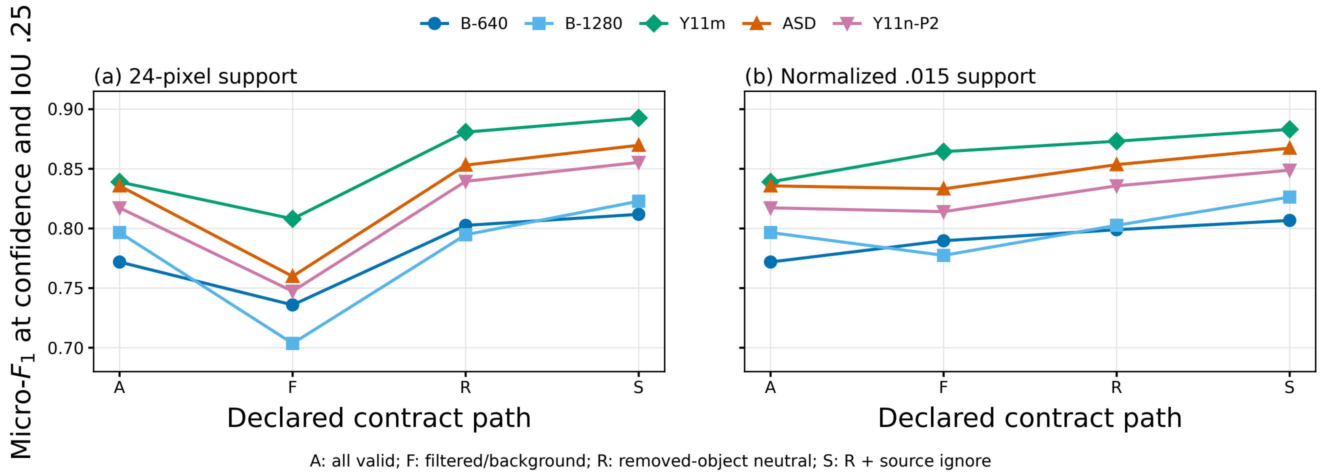

Fig. 2. Five-configuration VisDrone $A \rightarrow F \rightarrow R \rightarrow S$ paths at confidence and valid-match IoU .25. (a) Absolute 24-pixel support. (b) Normalized .015 support. Each line replays one frozen prediction artifact on the same 548 images.

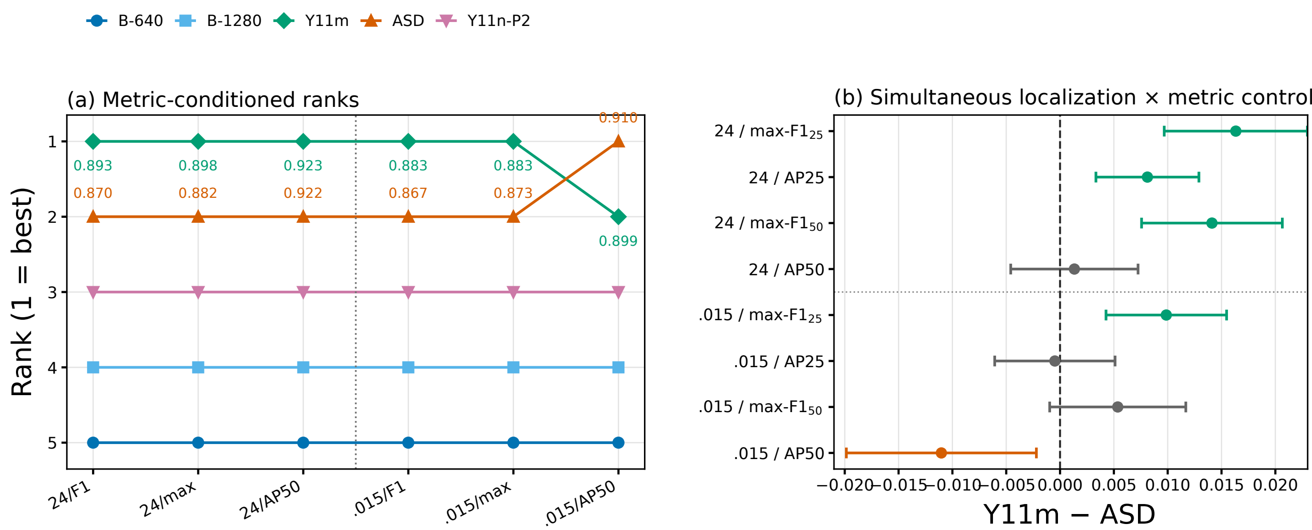

Fig. 3. Metric-qualified ranking under terminal contract S on the common 548-image VisDrone set. (a) Pool-wide ranks and Y11m/ASD point values. (b) Single-step max-t simultaneous 95% intervals for the eight Y11m–ASD max-F_1/AP comparisons; green, orange, and gray denote intervals above, below, and crossing zero. AP is recomputed from pooled detections in each of 10,000 paired resamples of 76 sequences.

design. The custom AP50 values match COCOeval at all four primary endpoints.

D. Evaluator controls

The declared AP cap is 2,000 detections per image, while the largest observed candidate output is 1,881; headline AP therefore contains no cap-truncated image. This matters because caps of 100 or 300 alter dense aerial-image comparisons even when the winner does not change. For the two candidates and two headline contracts, the independent score-ordered AP calculation and COCOeval agree to machine precision. A separate threshold-first Hungarian replay on the released matching-control subset agrees with greedy matching on the selected winner in all 24 multi-candidate settings; its largest absolute F_1 difference is .00546. The matching control validates an aligned component rather than extending the headline rank claim.

Image-paired controls preserve the same endpoint directions. Sequence intervals remain primary because adjacent VisDrone frames are not exchangeable independent observations; all image-level intervals are tabulated in the released records.

IV. Discussion

The gate, path, ledger, and cluster interval answer different questions. Coverage determines which population is observed. Retained support states which valid target remains. The ordered path attributes deterministic metric movement to one evaluator change at a time. Factorial metric controls and paired intervals then identify which metric factors change a named comparison on common sequences. A support shift is not sampling uncertainty, and a resolved F_1 margin is not automatically an AP conclusion.

We qualify each rank statement along three axes: common image coverage, the declared localization and score treatment, and the prespecified sequence unit. Fixed-confidence F_1 favors Y11m in all four pointwise intervals. Family-wise inference resolves three max-F_1 cells for Y11m, leaves normalized max-F_1 at IoU .50 open, and resolves normalized AP50 for ASD. The two retention-matched support definitions preserve these directional point patterns.

The cross-configuration replay distinguishes method portability from identical numerical behavior. The $F \to R$ mechanism is positive across all full-coverage VisDrone and UAVDT configurations, while its magnitude and the sign of $A \to F$ remain source and configuration dependent. Three AI-TOD resolution slices show the same positive removed-object step on the covered subset; they do not extend inference to the 935 images without predictions. The repeated mechanism across separately materialized VisDrone artifacts, UAVDT resolution and tiling configurations, and ancillary AI-TOD slices rules out dependence on one baseline cache.

The path is an audit coordinate system, not a claim that S is universally correct. Applications may choose whether removed valid objects should be background or neutral, but that choice should be explicit and its consequence reported. Likewise, resize invariance makes normalized support comparable under uniform scaling; it does not remove source semantics or acquisition differences. The three-source results empirically demonstrate both points.

The ordered path supplies information that an endpoint band cannot. Comparing A with S alone would merge a target change, a removed-object decision, and native ignore semantics into one number. The intermediate counts show which mechanism dominates and whether a recovery is caused by fewer FN, by neutralized FP, or by both. This attribution is especially important when a support cutoff removes many labeled objects, because a large score response can otherwise be mistaken for model instability or annotation error.

The practical output is a compact audit record rather than a preferred universal threshold. It contains the annotation denominator and prediction coverage; retained support in absolute and normalized coordinates; contract endpoints and integer count transitions; the exact common set and metric definition used for each comparison; and paired uncertainty at the source's dependence unit. These fields make two studies comparable even when they choose different valid application contracts. They also expose when an apparent robustness claim comes from a saturated detection cap, an incomplete prediction cache, a fixed confidence calibration, or image-wise resampling of correlated frames.

The evidence is tied to the released artifacts. The main AI-TOD evidence concerns full-annotation support and prediction coverage. Full-coverage contract evidence spans five VisDrone and four UAVDT configurations; VisDrone rank inference uses 76 filename-derived sequence clusters. The repository provides cross-configuration ledgers, simultaneous intervals, alternative-support controls, coverage diagnostics, and replay instructions: https://github.com/Marchematics/aerial-evaluation-audit.

V. Conclusion

Coverage-qualified contract auditing identifies the observed population, measures support, decomposes target and ignore semantics with exact count transitions, and separates localization, score integration, and sequence uncertainty in common-coverage ranks. Nine full-coverage configurations across two aerial benchmarks, ancillary AI-TOD resolution controls, simultaneous cluster intervals, and alternative-support definitions establish that the measured contract mechanisms are not unique to one prediction cache.

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